The Role of Academic Branding and Demographic Markers in AI-Driven Candidate Assessments: Exploring Bias in Automated Hiring Processes

Yufan Wu, Alex Mao, Gina Ge, Tiffany Yang

Emory University, Atlanta, GA

1. Introduction

Artificial intelligence (AI) technologies have rapidly entered the hiring landscape, especially in screening candidates for entry-level positions. While these tools promise efficiency and consistency, concerns persist about their fairness and potential biases. This research examines a core question in this evolving field: How do seemingly neutral factors—specifically, a candidate's academic department name and demographic markers—influence AI-based evaluations of suitability for an entry-level data science role?

This inquiry is not only timely but also locally relevant, as our institution's Quantitative Theory and Methods (QTM) department is considering rebranding its name. Understanding the relationship between department naming conventions and AI-driven assessments can guide institutional decisions and inform broader practices in academic and industry contexts. More importantly, this focus on textual and demographic cues highlights how subtle linguistic signals may create unintended disparities in automated hiring processes.

On a broader scale, the study advances current debates in ethical AI, algorithmic transparency, and the social implications of automated decision-making. By uncovering biases that may not be obvious to developers or users, the findings can help shape next-generation AI hiring tools

equipped with stronger safeguards. Ultimately, the insights gained here will deepen our understanding of how language and identity markers interact within algorithmic assessments, fostering more equitable and inclusive practices in the increasingly AI-driven employment landscape.

2. Related Research

The growing use of AI in hiring processes has raised concerns about algorithmic fairness and bias, particularly in resume screening and candidate evaluations. Existing research highlights the risks associated with algorithmic bias stemming from both explicit and implicit factors. For instance, Raji et al. (2020) emphasize that biases can emerge due to unrepresentative training data, flawed model architecture, or the operationalization of ambiguous hiring criteria. Their work underscores the need for rigorous auditing frameworks to mitigate biases in AI hiring tools (Raji et al., 2020). A related study by Binns (2018) delves into the interplay between algorithmic decision-making and social norms, arguing that while AI models may appear neutral, their outputs often reflect societal biases embedded in the data. This perspective is particularly relevant when considering how textual and demographic cues may influence hiring models (Binns, 2018).

Research has demonstrated that linguistic markers, such as the names of academic departments or the phrasing in resumes, can inadvertently shape AI evaluations. Studies by Caliskan et al. (2017) reveal that word embeddings commonly used in AI systems can encode stereotypes, leading to biased outputs even in seemingly neutral contexts. This finding is crucial for understanding how department names might influence AI judgments in hiring (Caliskan et al.,

2017). Furthermore, research by Holstein et al. (2019) identifies how AI models interpret textual inputs differently based on demographic cues, such as gendered language or cultural indicators. Their study provides evidence that even small variations in text can lead to significant disparities in AI assessments (Holstein et al., 2019).

3. Dataset

For datasets that better meet the criteria for our research question, we used OpenAI to generate three datasets with three input lists: department names (Data and Decision Sciences, Decision and Data Sciences, Quantitative and Data Sciences, Interdisciplinary Data Science, Integrative Data Science, Integrative Quantitative Theory, Quantitative Theory and Methods, Interdisciplinary Quantitative Sciences Statistics, Data and Decision Science, Interdisciplinary Data Sciences, Statistics and Data Science, Data Science), locations (US and International), and internships (2 possible statuses, “Yes” and “No”). Each dataset took one input list as a variable and held the other two constants. We generated 400 profiles in each dataset, for a total of 1200 profiles across all datasets. To conduct a better analysis of the data and discover potential patterns, we used those three datasets to generate another two datasets: one contains numerical ratings for each candidate based on their profiles and another contains qualitative ratings, specifying the strengths and weaknesses of each candidate.

The transition from the original dataset of candidate profiles to a dataset with numerical ratings was achieved through prompt engineering and the use of Azure OpenAI. The three original datasets of candidate profiles were merged using the pandas library to create a comprehensive dataset containing all profiles. A custom prompt was designed for each candidate with a request

for a numerical rating on a scale from 0 to 100, where 0 indicated a poor fit and 100 indicated an excellent fit for an entry-level data science position. Those prompts were used to query an Azure OpenAI model, simulating the role of an HR specialist and the model's output was validated to ensure no non-numeric or out-of-range responses. The dataset with qualitative evaluations was generated similarly: merging three datasets and using Azure OpenAI to evaluate the prompts. However, for some candidates, Azure OpenAI failed to generate strengths and weaknesses and responded with “This evaluation is not possible due to insufficient information”, which might influence the analysis process.

Certain limitations are exhibited in those two datasets. Qualitative evaluations of candidates’ profiles might introduce bias and inconsistencies in responses and consequently, more works are needed to conduct a valid analysis. Numerical ratings might oversimplify the dataset and fail to provide a holistic view of the role of those variables in the AI resume screening process. Furthermore, the dataset focuses on candidates in the data science area, limiting the possibility of generalizing the findings to other areas. Researchers must consider these limitations when formulating research questions and interpreting results. Ethics were considered when generating the dataset. If researchers use real information gathered from participants, all personal information should be anonymized to protect the privacy of participants. Researchers also need to gain informed consent for the use of the data and ensure that the data is only for legitimate research purposes. Therefore, to avoid those possible problems, we choose to use AI-generated data, which also aligns with the theme of our research.

4. Process and Methods

Our quantitative analysis aimed to understand how a candidate's academic department, location, and internship experience influenced AI-generated assessments, using methods drawn from computational social science workflows, including prompt engineering, dataset creation, exploratory data analysis (EDA), and statistical modeling.

We generated synthetic candidate profiles using Python's pandas library to ensure anonymity and experimental control. Three datasets were created with variations in department names (ten options), geographic location (U.S. vs. international), and internship experience (yes vs. no), then combined into a comprehensive pool of 1,200 profiles. This synthetic approach avoided privacy concerns and aligned with the study's focus on AI-driven hiring processes. Prompt engineering was used to query the Azure OpenAI model, simulating an HR specialist's assessment by generating numerical ratings (0–100). Clear instructions and a retry mechanism ensured valid responses. After data retrieval, cleaning procedures were applied to remove duplicates and filter invalid ratings, ensuring all values were within the 0–100 range. EDA methods, including histograms, boxplots, and bar charts (created with matplotlib and seaborn), visualized rating distributions and highlighted patterns by department, location, and internship status. These insights informed hypotheses about the influence of candidate attributes. To rigorously test these relationships, we employed Ordinary Least Squares (OLS) regression via statsmodels, incorporating categorical predictors and interaction terms. This allowed us to evaluate the combined effects of variables on AI assessments. While effective, the process had limitations. Numerical ratings oversimplified complex profiles, and reliance on a single LLM introduced potential biases. Advanced methods, such as transformers-based encoders, were considered but not fully implemented. Nonetheless, the chosen methods, such as LLM-based rating generation,

data cleaning, EDA, and regression modeling, were well-suited to the research question and provided a robust foundation for exploring how department names, locations, and internships influence AI-generated evaluations. These techniques offered a clear, interpretable framework and a stepping stone for future refinements.

Our qualitative analysis explored how academic department names and other attributes influenced the language used by the AI model in candidate evaluations. Using natural language processing (NLP) techniques, clustering, and lexical analysis, we identified thematic patterns in the AI-generated feedback. We collected qualitative feedback from the Azure OpenAI model, consisting of textual descriptions of strengths and weaknesses. After minimal preprocessing, such as stop-word removal, we applied TF-IDF vectorization to convert the responses into a numerical feature matrix. This method emphasized terms unique to individual responses while ensuring manageable dimensionality with a cap of 500 features. To uncover patterns, we used K-Means clustering to group responses into five clusters, each representing a distinct evaluative tone, such as positive feedback, vague commentary, or actionable suggestions. Thematic analysis of top TF-IDF terms in each cluster revealed defining vocabulary and conceptual themes. Word clouds were generated for each cluster using Python's WordCloud library, offering an intuitive visual representation of the prominent terms and thematic focus. Favorability was quantified by calculating the frequency of positive terms (e.g., "excellent," "qualified") within each cluster. By mapping departments to clusters and aggregating cluster favorability scores, we calculated a normalized favorability score for each department, linking linguistic patterns to the research question about the influence of academic branding. While effective, this approach had limitations. K-Means clustering provided interpretability but may not capture data complexity as

well as advanced methods like hierarchical clustering or LDA. Favorability scoring, based on positive term frequency, could benefit from sentiment analysis or custom lexicons. Additionally, the synthetic dataset and reliance on a single LLM limit the generalizability of the findings. By combining TF-IDF vectorization, clustering, and favorability scoring, we identified patterns in AI-generated qualitative feedback and linked these to academic department names, offering insights into how textual cues shape AI-driven evaluations.

5. Results and Discussion

Our primary goal was to understand how specific textual cues, namely, a candidate's academic department, geographic location, and internship status, might influence AI-driven assessments for entry-level data science positions. The results from both the quantitative (numerical ratings) and qualitative (textual evaluations) analyses provide valuable insights into this interplay. While certain patterns emerged, the findings also underscore the complexity and multifaceted nature of AI-based evaluations.

5.1 Dataset with numerical ratings

The distribution of the AI-generated numerical ratings, shown in Figure 1, reveals three distinct peaks at approximately 50, 70, and 80. Rather than a smooth continuum, the clustering of responses suggests that the AI model may be categorizing candidates into broad tiers of perceived suitability. This tiering could reflect subtle attributes in the profile prompts that the model internally ranks, even if no explicit criteria are provided. Such groupings highlight that the model's rating behavior may not be uniform and could be influenced by the textual cues embedded in the candidate's profile. A closer inspection by department name (Figure 2) shows

that while most departments have mean ratings clustered between 60 and 80, some departments display more variance. For instance, “Quantitative Theory and Methods” presents a broader spread of ratings, whereas “Statistics and Data Science” appears more consistently rated. This variation suggests that department naming conventions, possibly signaling differing emphases on theory, methods, or applied analysis, may shape the AI’s perception of candidate suitability. Although no single department uniformly received either the highest or lowest ratings, the differing spreads indicate that certain academic markers might be interpreted by the model in ways that influence the rating’s confidence or consistency. In terms of demographics and personal attributes, location proved to have little measurable impact. The mean ratings for candidates based in the U.S. versus those labeled “International” were nearly identical (approximately 65.7 vs. 65.3). This similarity suggests that at least in this synthetic dataset and with the chosen LLM, geographic origin alone does not drive significant differences in the AI’s judgments. Internship status, however, displayed a pronounced effect. Candidates with internship experience averaged ratings of about 76.8, while those without had a mean score of around 54.2. This substantial gap indicates that signals of practical work experience or applied skills, conveyed simply by the mention of having had an internship, strongly influence the AI model’s perceived suitability. From a hiring perspective, this finding may reflect how resume-screening models weigh experiential indicators more heavily than institutional labels or geographic cues. These quantitative findings contribute to our understanding of the research question. They suggest that while certain demographic cues (like location) may not be influential, subtle variations in department names and explicit indicators of professional experience (like internships) can meaningfully shift an AI model’s assessment. This complexity points to the importance of carefully considering what textual signals are embedded in candidate profiles, as

even ostensibly neutral academic identifiers or brief mentions of prior work experience can steer automated evaluations.

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Mean Ratings by Location:
 Location
International    65.300000
US               65.683333
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Table 1: Mean Ratings by Location

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Mean Ratings by Internship:
 Internship
No      54.216667
Yes     76.766667
```

Table 2: Mean Ratings by Internship

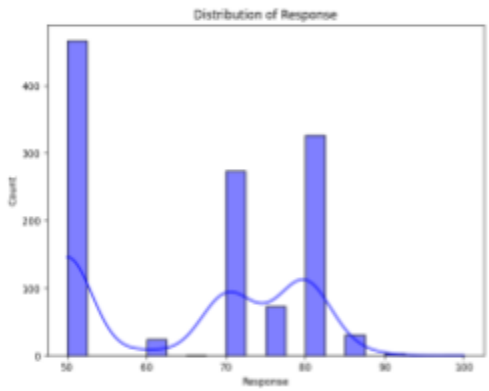


Figure 1: Distribution of numerical ratings

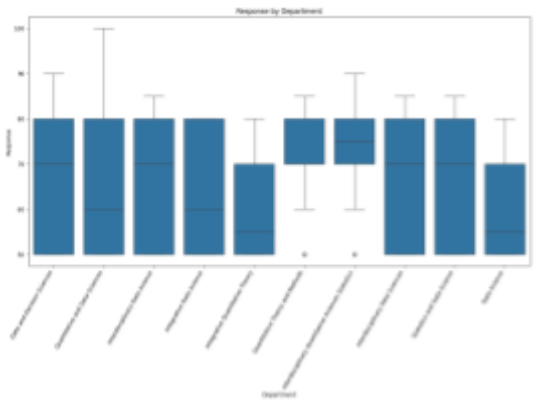


Figure 2: Response distribution by department

5.2 Dataset with qualitative evaluations

The clustering analysis uncovered five distinct evaluative tones:

1. Cluster 0: Specific Recommendations or Suggestions: This cluster was characterized by actionable feedback tied to complex or interdisciplinary department names. Common terms included "skills," "improvement," and "qualifications," suggesting detailed and constructive evaluations.
2. Cluster 1: General or Non-Specific Feedback: Feedback in this cluster was vague and often linked to unconventional or ambiguous department names. Terms like "insufficient

information" and "general observations" were dominant, reflecting a lack of specificity in the evaluations.

3. Cluster 2: Positive Evaluations: This cluster contained highly favorable feedback, with terms like "excellent," "strong," and "qualified" dominating. Departments with traditional, industry-aligned naming conventions, such as "Statistics and Data Science," were frequently associated with this cluster.
4. Cluster 3: Neutral or Ambiguous Responses: Balanced but non-committal feedback characterized this cluster. Terms like "average" and "moderate" were prevalent, often linked to composite or hybrid department names.
5. Cluster 4: Negative Evaluations or Limitations: Negative or incomplete feedback defined this cluster, featuring terms like "lacking" and "insufficient." Departments with unconventional or less interpretable names often fell into this group.

Figure3. Cluster 0: Specific Recommendations or Suggestions

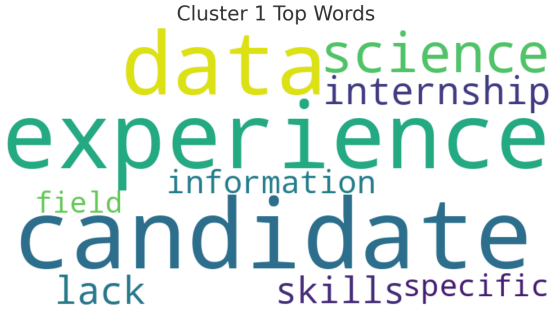
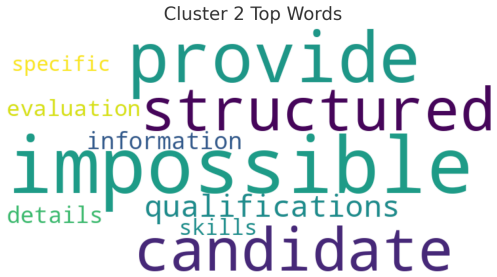


Figure4. Cluster 1: General or Non-Specific Feedback



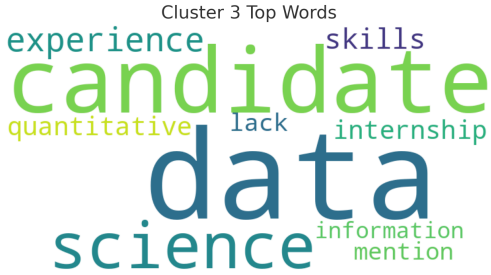


Figure5. Cluster 2: Positive Evaluations

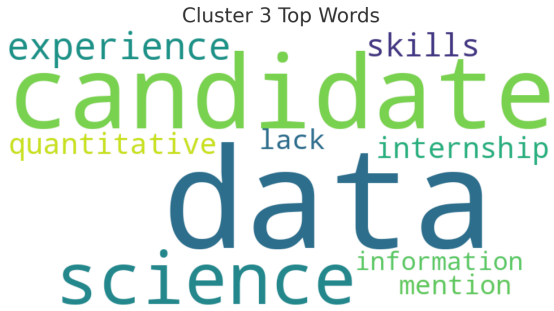


Figure6. Cluster 3: Neutral or Ambiguous Responses

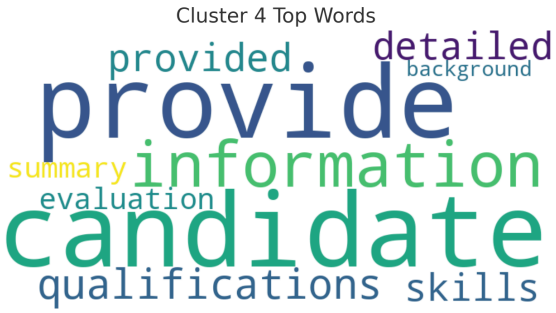


Figure6. Cluster 3: Neutral or Ambiguous Responses

Figure7. Cluster 4: Negative Evaluations or Limitations in the Response

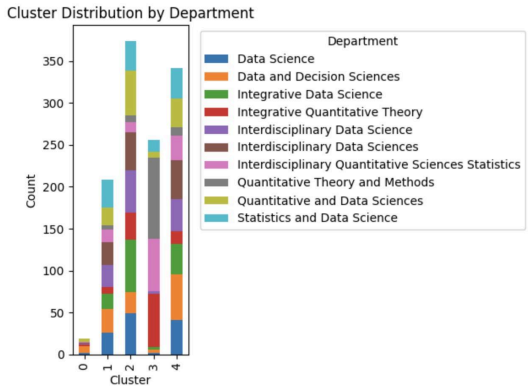


Figure8. Cluster Distribution by Department

These findings highlight the relationship between department naming conventions and the tone of AI-generated feedback. Departments with traditional, industry-aligned names elicited more confident and favorable feedback, while unconventional names prompted neutral or incomplete assessments. Internship experience also strongly correlated with positive evaluations, reinforcing the importance of professional exposure. Favorability was measured by analyzing clusters for positive language and feedback indicators. Terms such as "excellent," "outstanding," and "qualified" were used as proxies for positive feedback. By calculating the proportion of these terms within each cluster, a relative favorability score was assigned to each group. Cluster 2, representing positive evaluations, had the highest favorability score, aligning with departments featuring traditional, recognizable naming conventions. By mapping department names to their respective clusters and aggregating the data, normalized favorability scores were computed for

each department. Departments such as "Statistics and Data Science" consistently scored highly, while unconventional names ranked lower.

Department	0
Data Science	0.05969524302857640
Data and Decision Sciences	0.07789145011367230
Integrative Data Science	0.05474792697014920
Integrative Quantitative Theory	0.1428489206266980
Interdisciplinary Data Science	0.06754397865508980
Interdisciplinary Data Sciences	0.0703237092125981
Interdisciplinary Quantitative Sciences Statistics	0.1676695732251290
Quantitative Theory and Methods	0.1995424328757660
Quantitative and Data Sciences	0.06507032062587620
Statistics and Data Science	0.09466644466644470

Table3. Normalized Favorability Score by Department

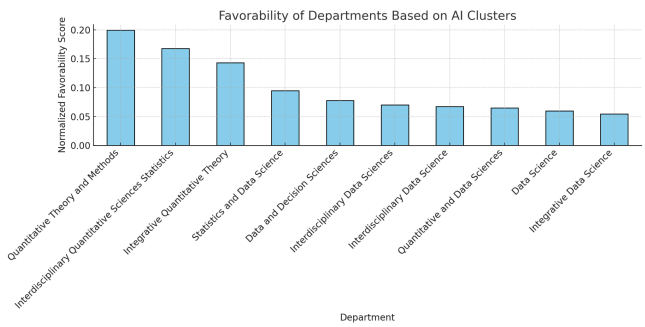


Figure9. Favorability of Departments Based on AI Clusters

6. Conclusion and Next Steps

In sum, our study illuminates the subtle yet impactful ways that AI-driven hiring tools interpret textual cues within candidate profiles. The findings suggest that not all department names are equally advantageous; those closely aligned with recognized fields, incorporating terms like “Quantitative,” “Theory,” “Methods,” or “Statistics”, tended to elicit more favorable assessments, both numerically and qualitatively. By contrast, departments with more novel or less conventional nomenclature prompted greater uncertainty and less detailed positive feedback, highlighting how even seemingly minor linguistic variations can influence automated

evaluations. Internship experience emerged as a particularly strong positive signal, raising candidate scores and prompting more specific, constructive commentary. Location, however, appeared to have negligible influence. These patterns underscore the importance of carefully considering the textual design of candidate profiles and academic branding to foster equitable AI-driven assessments. Despite these insights, challenges remain. Synthetic data, model variability, and the relative simplicity of our experimental conditions limit the generalizability of the results. Furthermore, qualitative evaluations occasionally lacked depth and consistency, pointing to the need for improved prompt engineering and more robust AI models. Looking ahead, expanding the dataset to include more diverse fields, attributes, and real-world data will help verify these preliminary findings. Collaboration with domain experts, implementing bias mitigation strategies, and conducting human-AI comparative studies will also be essential. Through such efforts, we can refine AI-based hiring systems to ensure that they support transparent, equitable, and informed recruitment processes while upholding ethical standards and fairness in decision-making.

Acknowledgments

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